

AMI - Auto Motivated Intruder: Celebrities re-identification using LLM

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1 Introduction

The burgeoning demand for expansive text data in social and medical research presents a challenge: while pressing questions necessitate using sensitive data, data protection and ethics raise a barrier. A promising solution lies in automated text anonymization, which performs data masking of Personally Identifiable Information ("PII"), such as names and locations, at scale. This work displays experiments on the topic, utilizing LLMs capabilities, as an off-the-shelf tool. The code and results can be found in AMI ¹.

Until today, numerous studies illustrated that text can be anonymized with only minor performance degradation. Nevertheless, this approach is not infallible, especially for sensitive sectors such as healthcare. How can we tell if our text is sufficiently anonymized? The domain of text anonymization lacks a standardized evaluation method, rendering the challenge multifaceted and open to individual interpretation. Traditionally, the most reliable evaluators have been humans attempting to re-identify individuals from anonymized texts, this process is called the Motivated Intruder Test. There are also some more technical metrics, such as the accuracy of tokens to be redacted, but again, this is prone to biases and personal interpretation.

Our aspiration is to surpass human re-identification capabilities, thereby setting a new benchmark for evaluating anonymized data. Namely, our approach can be used as an evaluation technique for other anonymization schemes, manual and automatic.

Our motivation is that a spontaneous re-identification process done by humans is indeed a threat, but in the world of Big Data, is not efficient and scalable. An automated version is very likely what will be developed and used by a potential Adversary.

Problem Statement:

1. **RQ1.** Compete with humans in the task of re-identifying famous persons from anonymized texts.

¹<https://github.com/golankai/AMI>

2. **RQ2.** Develop a new metric of the anonymization (/re-identification) rate of a single text.

We chose this problem as data collection and data protection are two very relevant and essential topics, and their convergence is not yet close to being fully discovered.

2 Related Works

2.1 Anonimization

DeID-GPT is "*a novel framework leveraging GPT-4 for automatic identification and removal of identifying information from medical text*" [4]. DeID-GPT preserved text structure and meaning, thus keeping downstream tasks' performance high while ensuring privacy. This tool uses Zero-shot prompting to an LLM, asking it to replace **PPIs** with *[redacted]* tokens.

Textwash [1] is an open-source tool, that addresses privacy in data sharing through semantic-preserving text anonymization. It utilizes supervised ML and fine-tuned BERT models to identify and redact 11 predefined categories of PIIs. Notably, it introduces "*potentially sensitive information*" ("**PSI**") for less defined identity-revealing text. Textwash ensures context-aware anonymization with its Transformer-based architecture. The process involves replacing sensitive content with tags and numeric suffixes for uniqueness, such as *DATE_1*.

2.2 Prompt Engineering

During the research, we were inspired by various SOTA **PE** methods and some we adopted accurately to measure the differences in their performance. Those methods have been shown to improve results on arithmetic, commonsense, and symbolic reasoning tasks, or some of them.

1. Recitation-augmented [2] which is a derivative of Generated Knowledge [6] says to ask the LLM to generate potentially useful information about a given question/prompt before generating a final response. We tested this approach similar to how the authors did in Self-Ask [5].
2. Self-consistency [7] is an approach that simply asks a model the same prompt multiple times and takes the majority result as the final answer.
3. Chain of Thought [9] ("**CoT**") prompting encourages the LLM to explain its reasoning as part of its answer, before concluding.
4. Multi-Persona Self-Collaboration [8] suggest upgrading the concept of Role Definition to the LLM, by instructing it to simulate several Personas to solve a task collaboratively.

3 Datasets

We used the datasets from the 2 studies in Textwash [1], which comprises person descriptions and their anonymized version. The descriptions are all of at least 500 characters and the anonymization was done by Textwash. Specifically, we focused on the famous persons there (e.g., Emma Watson, and Benedict Cumberbatch). In Study 1 in the original paper, there were 10 such persons with 401 descriptions. During the Intruder Test (Study 1), those descriptions were faced by the human "adversaries" 1200 times, leading to an average of 3 intruders per description. The dataset for Study 2 comprises 20 well-known individuals, with each individual represented by a collection of 49 to 61 text samples.

4 Algorithm - General approach

(**RQ1**) The primary objective of this study is to develop an **AMI** by leveraging the power of an LLM for the re-identification of famous persons from anonymized text data. We intend to use "*gpt-3.5-turbo*"², for its convenient access and proved great results. We hypothesize that these LLMs, due to their extensive training on diverse data sets, might possess the capability to accurately identify these well-known individuals, given the significant amount of publicly available information about them. We are curious to check the overall performance of LLMs in this task while exploring different PE techniques as mentioned in subsection 2.2. We are keen to learn the difference between them and their impact on the overall performance.

(**RQ2**) The first question is what is a good grade/score for an individual anonymized text? To our knowledge, there is no generic metric to evaluate the re-identifiability property, but more technical scores such as F1-score on annotated anonymization data or by a human Motivated Intruder Test [1]. We conjecture that the human re-identification rate is general, scalable, and can serve as a good metric. We thus define our gold labels of each text to be its re-identification rate, calculated as the normalized success rate of the humans. We first want to fine-tune a RoBERTa model [3] in the circumstances of low data volume availability to predict this metric. We will then do a similar exploration process of PE as for RQ1 to try optimizing the results derived by ChatGPT.

5 Experiments

During the research, we have tried different PE methods, as described in subsection 2.2. We manually optimized them by changing the order of the prompt's parts and experimenting with different phrasings.

²<https://chat.openai.com/>

5.1 Re-identification

5.1.1 Setup

We employed *gpt-3.5-turbo* as a simulated intruder for conducting experiments on the Study 2 dataset. We selected 30 text samples attributed to Adele for pre-training and prompt rephrasing across different PE techniques. Our experimental framework includes comparisons across three well-known individuals, totaling 166 samples. These individuals—Adele, Craig, and Jagger—were re-identified by human evaluators 38%, 22%, and 9% of the time, respectively. Ultimately, we applied the most effective prompt methodology to the entire dataset and compared its performance against human evaluations, utilizing 1,080 samples. Each experimental process employs a distinct PE technique.

5.1.2 Experiments

Our experimental design consists of five distinct processes. The first two, labeled as "*GK-one*" and "*GK-multi*", employ the Generated Knowledge technique. The "*GK-one*" approach utilizes a single, comprehensive query, while "*GK-multi*" involves multiple, refined queries for a more nuanced analysis.

The third process, "*completion*", employs text completion techniques, which have previously demonstrated efficacy in enhancing performance on certain tasks.

The fourth and fifth processes, termed "*zero-shot*" and "*zero-shot-conf-guess*" respectively, employ zero-shot learning methodologies, accompanied by task-specific explanations. The fourth process solely provides an explanation of the task, while the fifth incorporates a conditional guessing mechanism based on confidence levels.

Evidently, the fifth process outperforms the others, although the underlying reasons for this are not conclusively understood. It appears that allowing the LLM to operate with a degree of confidence in its knowledge base may be beneficial, particularly when sufficient data is available, as is the case with well-known individuals.

Table 1: PE comparison - re-identification successful rate on Adele, Craig and Jagger

	human rate	GK-one	Gk-multi	completion	zero-shot	zero-shot-conf-guess
Adele	0.38	0.18	0.24	0.18	0.27	0.76
Craig	0.22	0.04	0.07	0.02	0.11	0.27
Jagger	0.09	0.13	0.13	0.14	0.13	0.43

5.1.3 Humans Vs LLM

Table 2: Comparative Analysis of Human and Language Model Performance in Zero-Shot Re-Identification with Confidence-Based Guessing by *gpt-3.5-turbo*

	Human	Exp-1	Exp-2		Human	Exp-1	Exp-2
Adele	0.389	0.746	0.729	Hamilton	0.508	0.918	0.902
Bale	0.098	0.412	0.353	Jagger	0.094	0.491	0.472
Beckham	0.551	0.878	0.857	John	0.389	0.889	0.889
Campbell	0.196	0.333	0.314	Middelton	0.446	0.804	0.839
Craig	0.222	0.315	0.277	Moss	0.228	0.562	0.561
Cumberbatch	0.074	0.204	0.204	Radcliffe	0.102	0.559	0.593
Delevigne	0.235	0.333	0.356	Rowling	0.294	0.804	0.824
Dench	0.054	0.268	0.250	Sheeran	0.450	0.700	0.750
Gervais	0.283	0.391	0.370	Smith	0.161	0.554	0.589
Grant	0.229	0.521	0.542	Watson	0.241	0.778	0.796
				Overall	0.263	0.578	0.579

In a comprehensive evaluation of re-identification performance, humans exhibited an overall success rate of approximately 0.263. In contrast, the language model gpt-3.5-turbo demonstrated significantly higher efficacy with scores of 0.578 and 0.579 in Experiment 1 and Experiment 2, respectively. Moreover, the language model maintained a consistent performance across multiple runs, exhibiting only marginal fluctuations in success rates. Utilizing self-consistency techniques appears promising for further enhancing these results, a trend observed across all experimental processes. Additionally, the performance gap between the language model and human success rates remains proportionally consistent across individual subjects. This suggests that the language model’s superior performance is not restricted to specific individuals but rather represents a consistent advantage over human capabilities in re-identification tasks.

5.2 Single-text Grading

As we wanted to compare the PE experiments with the model’s fine-tuning, we split the data such that the validation and test datasets contained 40 samples each, from study 1 and only of famous persons.

To evaluate the quality of each experiment (models/prompts) we looked at 3 metrics: (1) the Root Mean Square Error (“**rmse**”) that measures “how far” the predictions are from the true values; (2) Average Prediction, as a simple indication if our “model” is over/under predicting; (3) Spearman Correlation to assess how well the predictions are correlated with the truth values.

For the fine-tuning of the model, we tried different configurations, including data augmenting with more examples of the Semi-famous persons or the fictive ones [1]. We tried training the classifier layer only or the 11th encoder layer as well. Overall, we found that the best model was trained with data from both studies, using only the famous persons and training both the classifier and the

Table 3: Re-identifiability scoring results on the 40-sample test set.

	rmse	avg_pred	spearman
human rate (gold)	0	0.42	1
RoBERTa	0.41	0.42	0.32
zero-shot	0.53	0.62	-0.22
multi persona	0.44	0.43	0.24
one-shot 0	0.46	0.52	0.12
one-shot 1	0.45	0.54	0.17
three-shot	0.44	0.36	0.26
CoT	0.55	0.81	0.39
self-consistency with zero-shot	0.48	0.65	0.17
self-consistency with three-shot	0.42	0.36	0.23
Roles mean	0.55	0.79	0.22
Role 1	0.59	0.73	-0.06
Role 2	0.6	0.83	0.15
Role 3	0.57	0.81	0.35
Role 4	0.57	0.78	0.21

11th layer. This model achieved a rmse of 0.38 and a Spearman Correlation score of 0.25 on the validation set.

We then compared the PE techniques’ overall scores with the ones of the fine-tuned model. The approaches we checked and their results can be found in Table 3. Notably, we looked at the following approaches:

1. Multi Persona [8] - We examined the difference between asking 4 different prompts and calculating the average score between them, "Roles" in the table, and asking the LLM to simulate the 4 at the same time. We noticed that There was not too much of a difference in their performance, though the distributed method (mean of roles) was more stable across runs.
2. Self-consistency [7] - We tried 2 types, one with 0-shot prompts and one with 3-shot, the latter had slightly better results, which aligns well with the fact that the normal 3-shot strategy also had better results than both of the 1-shot ones.

The seemingly best performers were the RoBERTa model and CoT prompt strategy, as can be seen in Table 3. To further examine, we show in Figure 1 the distribution of their results with the human rates (our gold values). One can see that both indeed show some trend, but not very strong. Similar, mostly worse distributions have been seen for the other PE methods.

A general result we have seen is the different performance for runs of the same experiment, for example, in the grading part, we ran "Role 4" twice, one resulted in a spearman correlation of 0.6, the other 0.29. We thus, re-ran many of the experiments which had really bad results, and with most, we could get better scores.



Figure 1: RoBERTa and CoT scoring distribution

Noticing that CoT, Role 3, and some more methods had a rather high rmse and not bad Spearman score, we inferred that these models are over- or under-shooting, but still follow the trend. We thus decided to manually tweak those predictions, to "fix" them, by subtracting or adding some α to each prediction. Aligned with the distributions in Figure 1 and similar visuals, this method is not very effective. It indeed decreased the rmse scores, so the models are mathematically better, but in practice, they didn't fit the data much better. For example for for CoT with $\alpha = -0.4$, we got rmse of 0.39 and a Spearman Correlation score of 0.38. But again, this is only seemingly better and not practical.

6 Conclusions

This project provided some interesting insights. Generally, we noticed how subtle PE is and how a minor, seemingly unimportant, rephrasing can drastically alter the results. For example, in **RQ1**, while exploring the zero-shot method, we noticed that changing the phrase *re-identify* to *guess* (based on confidence level) resulted in surprisingly better results and use both. We also learned that in the field of PE, it is necessary to experiment with different methods. Another key point was the variance of the LLM's answers, we noticed that when dealing with very short answers, such as one name or a score, even with low temperature, the answers were rather inconsistent. This led to very similar runs with very different results and further emphasizes the importance and strength of Self-consistency [7].

More specifically, we believe now that LLMs can enrich the capabilities of an adversary when dealing with data anonymization. We propose AMI as a baseline for such a task but are also aware that this AMI is very specific for the context of famous persons' general descriptions. We are curious about ex-

panding this concept to other fields, such as medical data. These study findings bolster the case for the strategic deployment of advanced language models in re-identification tasks, while also highlighting their potential role in formulating robust evaluative benchmarks. However, in the experiment of anonymization grading our success was limited. We thus infer that LLMs were not a suitable tool for the task and that given such a small amount of data, fine-tuning is also not very successful. Another option to consider is that our truth values, based on the success rate of human intruders are not consistent enough to form a correlation to be learned. To further examine these claims we suggest repeating the experiments with either more data or more adversaries per description which will result in a more smooth distribution of the god labels.

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